Ternary Operator in C

1. One-Line Calculator

You are developing a simple calculator for a mobile application. The user enters two integers and an arithmetic operator (**+, -, *,** or **/**). Write a C program to perform the selected arithmetic operation using the **ternary operator**.

If the user enters an operator other than **+, -, *,** or **/**, display **"Invalid Operator"**.

Example Input

```
Enter first number: 20
Enter operator: *
Enter second number: 5
```

Example Output

```
Result = 100
```

2. Secret Door Game

You are creating a simple adventure game where a player encounters three mysterious doors. The player must choose a door by entering **1, 2,** or **3**.

Door 1: The player finds a treasure.
Door 2: The player encounters a ghost.
Door 3: The player finds the exit.

If the player enters any other number, display **"Invalid Door"**.

Write a C program to determine the outcome using the **ternary operator**.

Example Input

```
Choose a door (1-3): 1
```

Example Output

```
You found the treasure!
```

3. Elevator Floor Detector

You are developing a smart elevator system for a building. The user enters a floor number, and the system should display the appropriate description.

Rules:
• 0 → Ground Floor
• Negative number → Basement
• 1 → First Floor
• 2–12 → Regular Floor
• 13 → Unlucky 13th Floor
• Greater than 13 → Upper Floor

Write a C program to determine the floor description using the **ternary operator**.

Example Input

```
Enter floor number: 13
```

Example Output

```
Unlucky 13th Floor
```

4. Detect a Triangle Type

You are developing a geometry application that identifies the type of triangle based on the lengths of its three sides. The program should accept three positive integers **a, b,** and **c**.

First determine whether the three sides can form a valid triangle.

Triangle Validity Formula

```
a + b > c
b + c > a
a + c > b
```

If the triangle is valid, classify it as:
• **Equilateral Triangle** → All three sides are equal.
• **Isosceles Triangle** → Any two sides are equal.
• **Scalene Triangle** → All three sides are different.

If the sides cannot form a triangle, display **"Not a Triangle"**.

Write a C program using the **ternary operator** instead of **if-else**.

Example Input

```
Enter three sides: 5 5 8
```

Example Output

```
Isosceles Triangle
```